pfi

Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

AdaBoost

WEEK 2 · DAY 1

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Side by side, or one after another

	Bagging and forests (week 1)	Boosting (week 2)
Rows each tree sees	its own bootstrap sample	every row, with weights
Order	independent	one after another
The trees	deep and unstable	small and weak: stumps
Combined by	a plain average	a weighted vote
What it fixes	the noise of one deep tree	the bias of one small tree

Each new tree is aimed at the rows the trees before it got wrong.

A stump: a tree with one question

The stump asks

```
PAY_0 <= 1.5
```

16.51%

default rate, yes side (16,115 rows)

70.13%

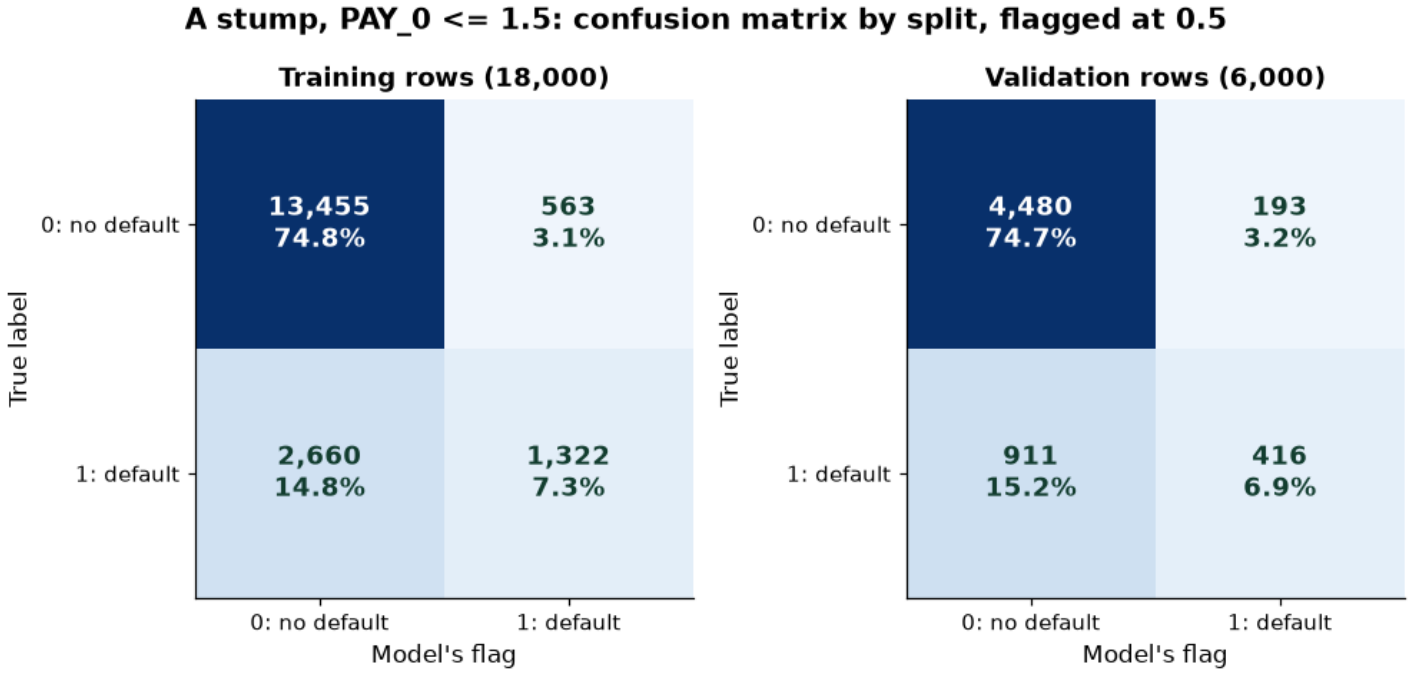
default rate, no side (1,885 rows)

0.6361

validation ROC AUC

Weak on purpose. AdaBoost uses two hundred of them.

The standard view: one heatmap per split



One question cannot memorise rows: the two panels look alike.

How to read it

True label down the side, the model's flag at 0.5 across the top. Each cell: a count and its share of that split's rows.

3,223 off the diagonal

The training panel's two wrong cells are the rows round one will reweight.

Round one by hand

1. Fit

a stump on the weighted rows

2. Error

share of the weight on missed rows

3. Weight

ln((1 - e) / e), samme_alpha

4. Reweight

missed rows times e^alpha, then / sum

3,223

rows missed of 18,000

0.1791

weighted error e

1.5228

stump weight

Every row starts at 1/n, so round one's weighted error is the share of rows missed.

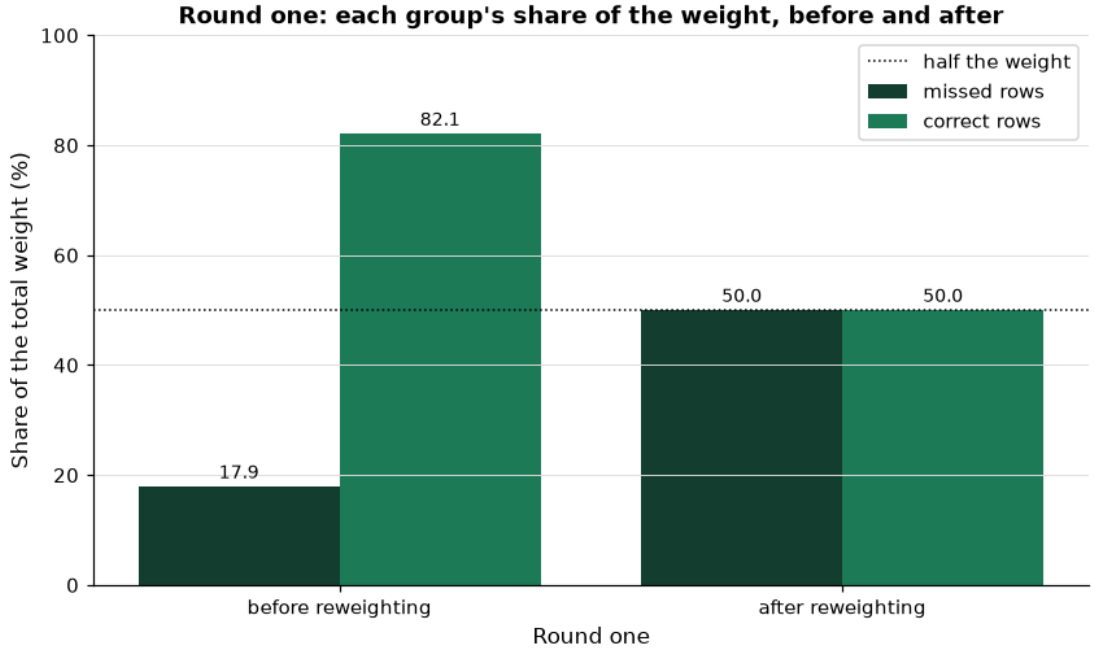
After reweighting, the missed rows carry half the weight

0.1791
missed rows' weight, before

0.5000
after

4.5849
missed over correct

```
e  x  (1 - e) / e  =  1 - e
the missed rows now weigh what the correct rows weigh
```



On the new weights the last stump is a coin toss. The next stump has to find something else.

Three rounds by hand, matched to scikit-learn

Round	Question	Weighted error	Hand weight	scikit-learn
1	PAY_0 <= 1.5	0.1791	1.5228	1.5228
2	PAY_AMT1 <= 1223.5	0.4128	0.3526	0.3526
3	PAY_3 <= 1.5	0.4356	0.2592	0.2592

Equal within 1e-12. AdaBoostClassifier is these four steps, repeated.

```
ada_3.estimator_weights_     # each stump's weight in the vote
```

Two hundred stumps

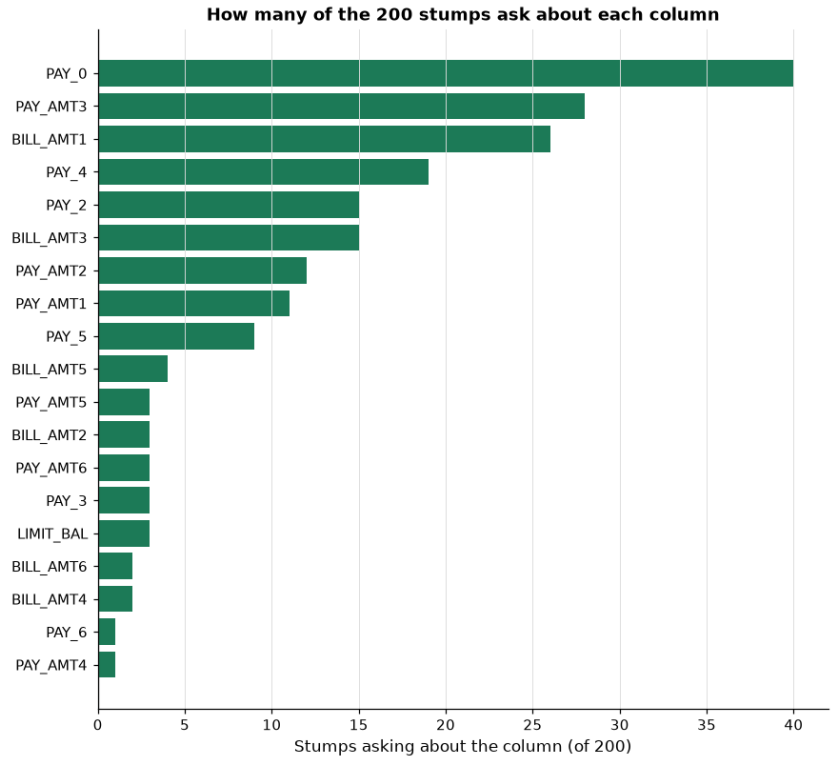
0.7708
validation ROC AUC

0.5321
average precision

0.6361
one stump

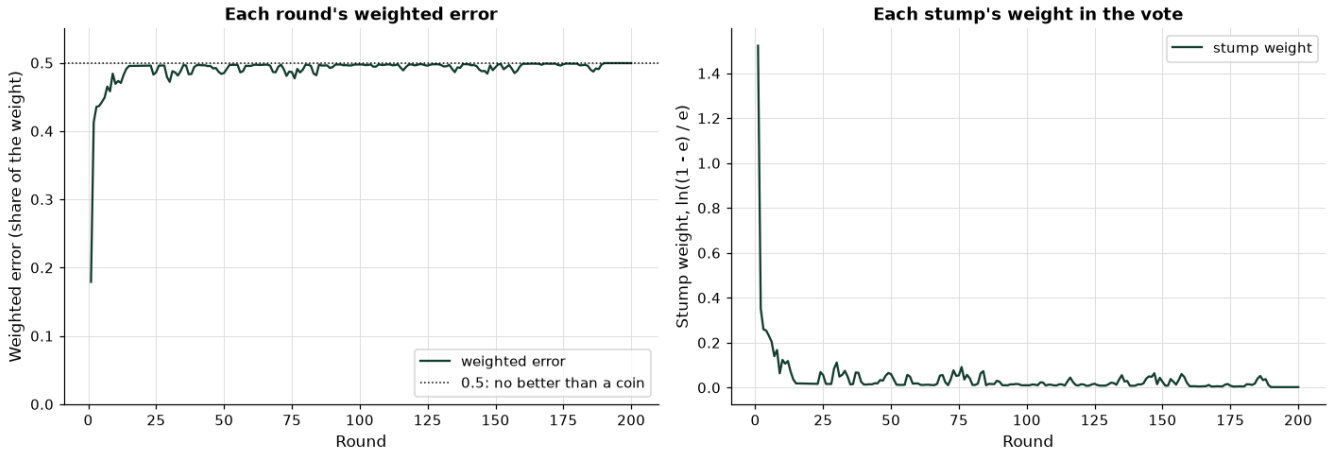
The stumps ask about 19 of the 19 columns

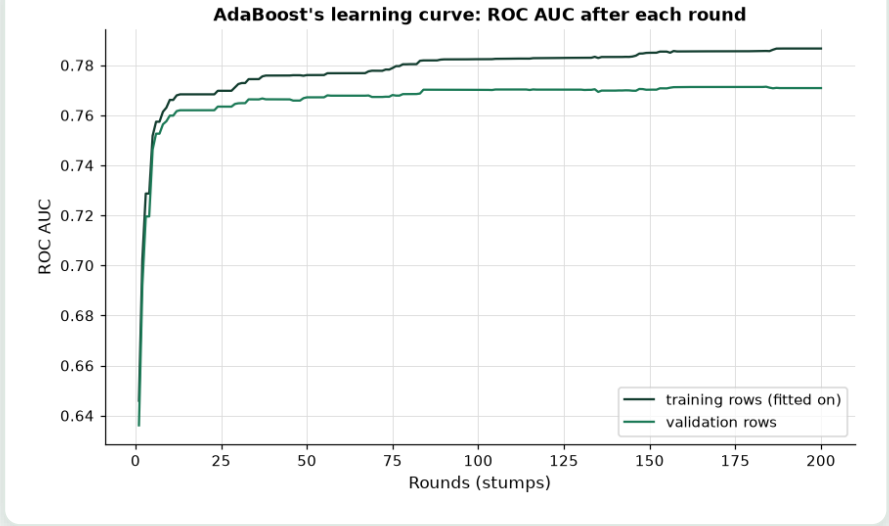
```
PAY_0 40 times
PAY_AMT3 28 times
BILL_AMT1 26 times
```



A sum of one-column steps: it bends along each column, never two at once.

Round by round: the error, the weight, and the learning curve





0.7867

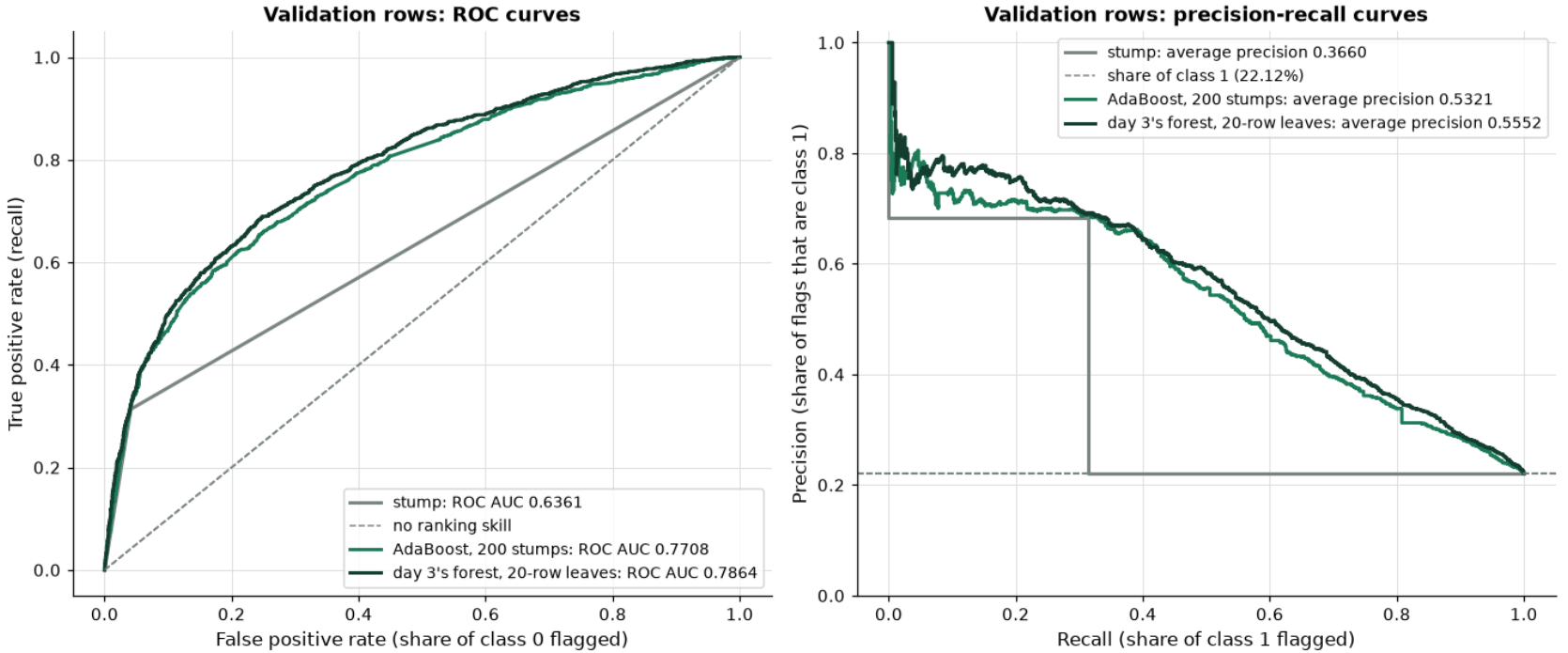
training rows, after 200 rounds

0.7708

validation rows

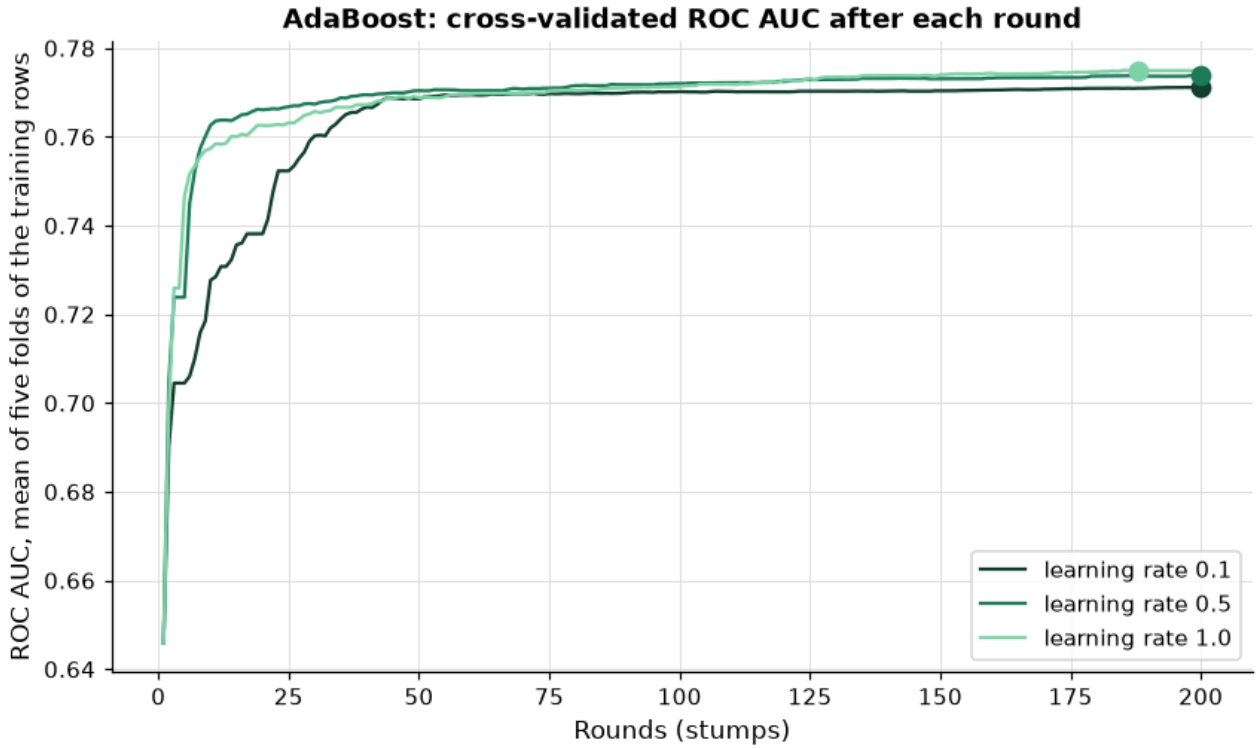
Later stumps are barely better than a coin and get a small say. The two curves stay close: little is memorised.

AdaBoost beside day 3's forest: validation rows



Forest 0.7864 ROC AUC, 0.5552 average precision; AdaBoost 0.7708 and 0.5321. The stump's curve has one corner.

The learning rate and the number of rounds



Rounds	0.1	0.5	1.0
10	0.7277	0.7626	0.7574
50	0.7686	0.7704	0.7688
200	0.7712	0.7738	0.7749

Rate 1.0 peaks at round 188, 0.7750. Five folds of the training rows; the validation rows play no part.

The learning rate mostly decides how many rounds it takes, not where the curves end.

A weighted vote, not a frequency

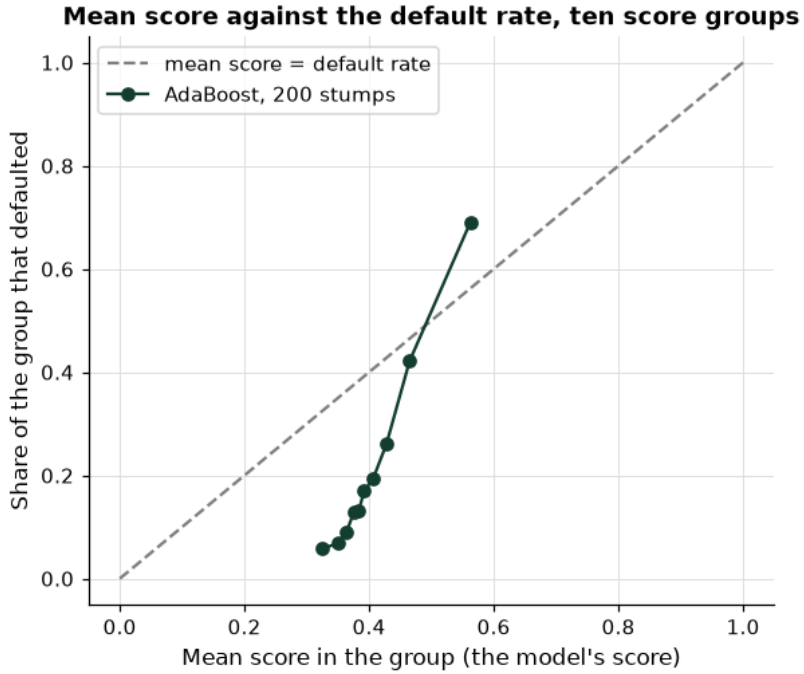
0.1192
lowest possible

0.8808
highest possible

0.4051
mean output

0.2212
default rate

Score group	Mean score	Default rate
1	0.3244	0.0583
5	0.3832	0.1308
10	0.5623	0.6900

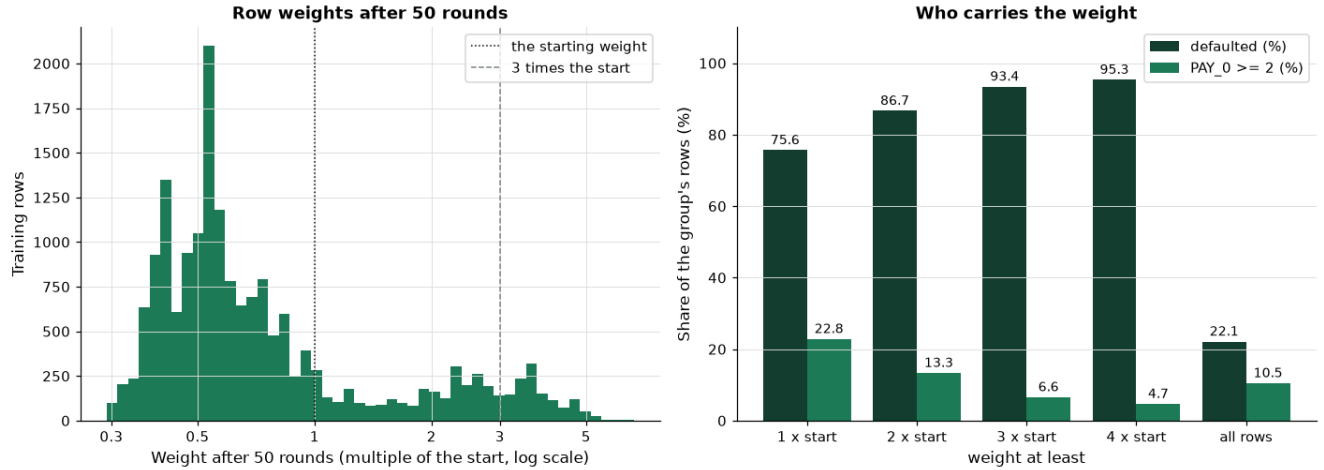


It ranks well; its level is not a share of defaults. The model's score, never a probability. Vote -0.4803 to 0.3243.

The rows that carry the weight after 50 rounds

Weights run from 0.29 to 6.63 times where they started. The rows at three times or more:

Rows	Count	Share of weight	Default rate	PAY_0 >= 2
Weight 3 x start or more	1,309	27.8%	93.4%	6.6%
All training rows	18,000	100.0%	22.1%	10.5%



Mostly accounts that defaulted while their latest status looked current: where the model struggles in this file, not a judgment of the accounts.

Where the week's name comes from

Yoav Freund and Robert E. Schapire, August 1997

"A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting"

Journal of Computer and System Sciences, volume 55, issue 1, pages 119 to 139

The last word of the title is the name of this week.

Source, read 5 October 2026: the Crossref record of the paper (DOI 10.1006/jcss.1997.1504). Bibliographic facts only; the paper itself was not read for this course.

Excel side by side

Excel side by side (tested in Excel)

`=1/ROWS($A$2:$A$18001)`	starting weights, D2 down
`=SUMPRODUCT(D2:D18001,A2:A18001)/SUM(D2:D18001)`	weighted error 0.1791
`=LN((1-M2)/M2)`	stump weight 1.5228
`=D2*EXP($M$3*A2)`	missed rows grow, E2 down
`=E2/SUM($E$2:$E$18001)`	new weights, F2 down
`=SUMPRODUCT(F2:F18001,A2:A18001)`	missed rows' share 0.5000
`the same on F and H with B and C`	rounds 2, 3: 0.3526, 0.2592

No AdaBoost in Excel; the stumps' missed columns come from Python. LOG is base 10: =LOG(4) gave 0.6021, =LN(4) gave 1.3863.

Save the day: git stash

1. Commit

mlkit.py, ensemblekit, tests

2. Log

results.csv at the out-of-fold threshold

3. Stash

a learning-rate experiment, put aside

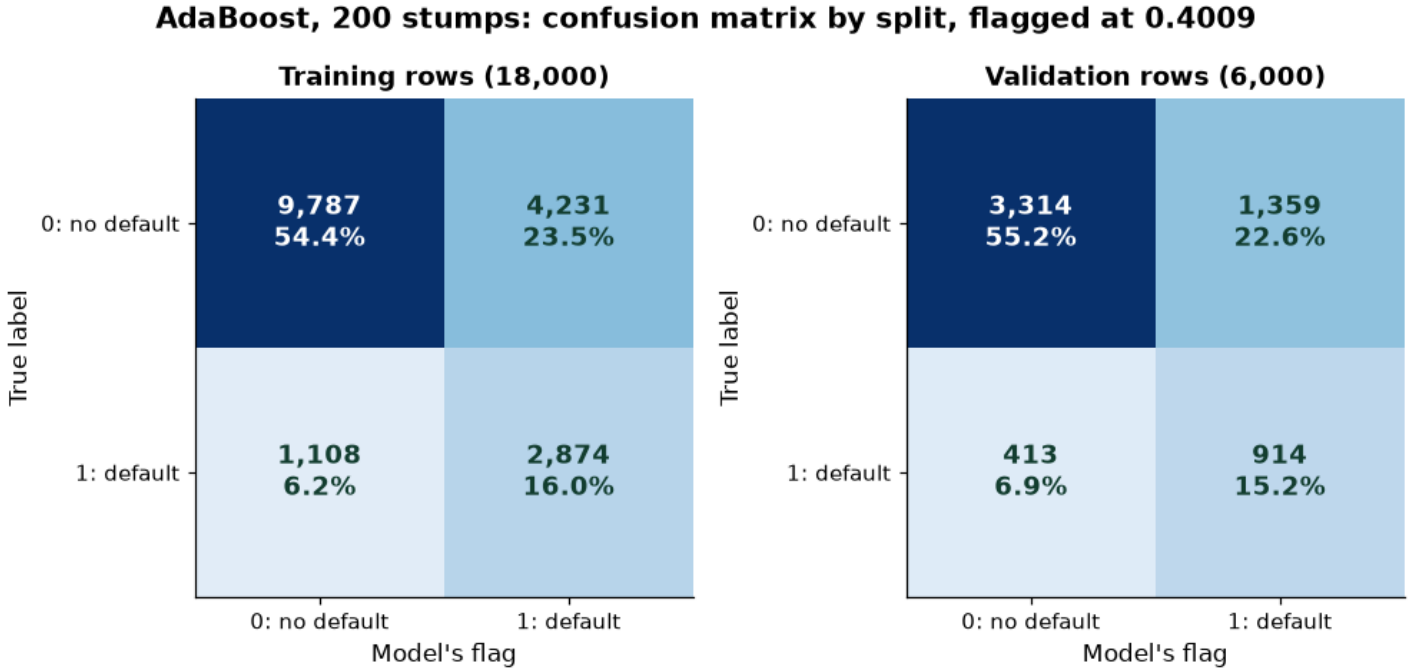
4. Drop

compare, then throw it away

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
adaboost_200_stumps,validation,0.7708,0.5321,0.4009,0.7047,0.4021,0.6888,0.5078
+adaboost_200_stumps_rate_0.5,validation,0.7706,0.5327,0.3164,0.6980,0.3955,0.6918,0.5033
```

Threshold 0.4009, from out-of-fold scores, far from 0.5. Rate 0.5 gives 0.7706: stashed and dropped.

The standard view at the out-of-fold threshold



Rows	Recall	Precision
training	0.7217	0.4045
validation	0.6888	0.4021

Recall on the training rows and on the validation rows sits close: the panels look alike, as the learning curve said.

Flagged at 0.4009, from out-of-fold scores on the training rows, never at 0.5.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which rows fit, which rows choose, which rows report: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the stump's weight is computed with np.log10: positive, plausible, and a learning rate nobody chose.

Code that runs is not the same as code that is right.

What you can do now

- Boosting: trees one after another, a weighted vote
- Run a round by hand: error, samme_alpha, reweight
- The missed rows end with half the weight
- Compare settings inside the training rows
- AdaBoost's output is the model's score

Week 2, Day 2 Gradient boosting from residuals, on the benchmark bonds